MF KVM v2 r7f 電流與壓降預算

來源：v2_r7f/tools/r7/budget_r7.py 對 v2_r7f/out/p4_kvm_mini.kicad_pcb 重跑的輸出（計算程式輸出，原文英文，照原樣排版）

負載電流沿用 r6 / r7 的預算。這支工具沒有把 U16 eFuse 算進去：U16 RON 最大 47 mΩ，全負載 1.51 A 時約 0.11 W（I²R，計算值）。外接風扇用自己的 USB 供電，不在這份預算內。

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V1V2 load 182.3 mA typical, 273.5 mA worst (AVDD12 + VDDC + VDD_MIPI)

Copper, solved on the board (25 C):
  V1V2 L2.2 to R65.2 (sense point), power path                  92.9 mOhm
  V1V2 R65.2 to FB3.1                                           11.9 mOhm
  V1V2 R65.2 to FB4.1                                            3.3 mOhm
  V1V2 R65.2 to FB5.1                                            9.8 mOhm
  V1V2_SNS R62.1 to R65.1 (carries the divider current only)   362.4 mOhm
  V2V5 U4.5 to FB2.1                                            36.7 mOhm
  resistors 1%: sense point 1.1725 to 1.2486 V
  resistors 0.1%: sense point 1.1831 to 1.2373 V

U15 at the sense point: nominal 1.2102 V, worst 1.1831 to 1.2373 V
  AVDD12 at the ball, typical current, 25 C, DCR 0.045: drop 9.4 mV -> 1.1737 to 1.2373 V  (limit 1.15-1.25, T
oshiba p15)
  AVDD12 at the ball, worst current, 70 C copper, DCR 0.09 (after test): drop 21.4 mV -> 1.1617 to 1.2373 V
(limit 1.15-1.25, Toshiba p15)
  VDDC: at least 1.1331 V before the ball (50 mV allowed for bead and copper; r6 measured 68-83 mOhm) against
1.1 V
  VDD_MIPI: at least 1.1331 V before the ball (50 mV allowed for bead and copper; r6 measured 68-83 mOhm) agai
nst 1.1 V
  R66 100R fallback, both paths intact: sense node 0.110 mV toward U15 at worst current (included in the windo
w)
  If the remote sense opens (R65, the line): U15 regulates at its output through R66; the power path drop adds
30.4 mV at worst current: AVDD12 down to 1.1331 V. Out of spec but no overvoltage
  R61 or R62 open: not covered by R66; R63 pulls FB to 0, U15 goes to 100% duty and V1V2 follows 3V3. R63 ope
n: FB follows the output, V1V2 about 0.6 V (safe)

U15 typical: Iout 182.3 mA, efficiency 0.88 ASSUMED (the AN p5 curves are for 5 V in), Iin 76.0 mA from 3V3, l
oss 30 mW in U15 + L2
  U15 junction rise if all the loss were in U15: 5 C (theta JA 170 C/W, AN p2); U5 was +54 / +81 C
  3V3 rail 546 mA (r6 652): U4 now carries only AVDD25 12.8 mA instead of 195.2 mA
  5V in 385 mA (r6 about 460), U2 at 27% of its 2 A

U15 worst: Iout 273.5 mA, efficiency 0.85 ASSUMED (the AN p5 curves are for 5 V in), Iin 124.4 mA from 3V3, lo
ss 60 mW in U15 + L2
  U15 junction rise if all the loss were in U15: 10 C (theta JA 170 C/W, AN p2); U5 was +54 / +81 C
  3V3 rail 1312 mA (r6 1461): U4 now carries only AVDD25 19.2 mA instead of 292.8 mA
  5V in 1019 mA (r6 about 1134), U2 at 66% of its 2 A

U4 TLV73325: 19.2 mA of 300 mA (6%), dissipation 18 mW, rise 4 C (RthJA 228.4, SBVS235 p5). r6: 293 mA (98%)
ripple: L 2.20 uH, VIN 3.3 V: inductor 348 mA p-p, peak 448 mA; on 44 uF 1.0 mV p-p (capacitive part only; ESL
spikes and light-load pulse skipping not modelled)
ripple: L 2.20 uH, VIN 3.3 V: inductor 348 mA p-p, peak 448 mA; on 30 uF 1.5 mV p-p (capacitive part only; ESL
spikes and light-load pulse skipping not modelled)
ripple: L 1.76 uH, VIN 3.4407 V: inductor 446 mA p-p, peak 496 mA; on 44 uF 1.3 mV p-p (capacitive part only;
ESL spikes and light-load pulse skipping not modelled)
ripple: L 1.76 uH, VIN 3.4407 V: inductor 446 mA p-p, peak 496 mA; on 30 uF 1.9 mV p-p (capacitive part only;
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ESL spikes and light-load pulse skipping not modelled)
AVDD12 limit is 40 mV p-p (Toshiba p15 VSN12); FB3 and C15/C16 filter further. Not measured.
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